UNIVERSIDAD INTERNACIONAL DEL ECUADOR

Information Systems Engineering



Connecta

Phase 2: Packet Tracer Network Prototype

Network Communication Technologies

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Repository URL: https://github.com/ashweeps/connecta-chat-empresarial

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1. Executive summary

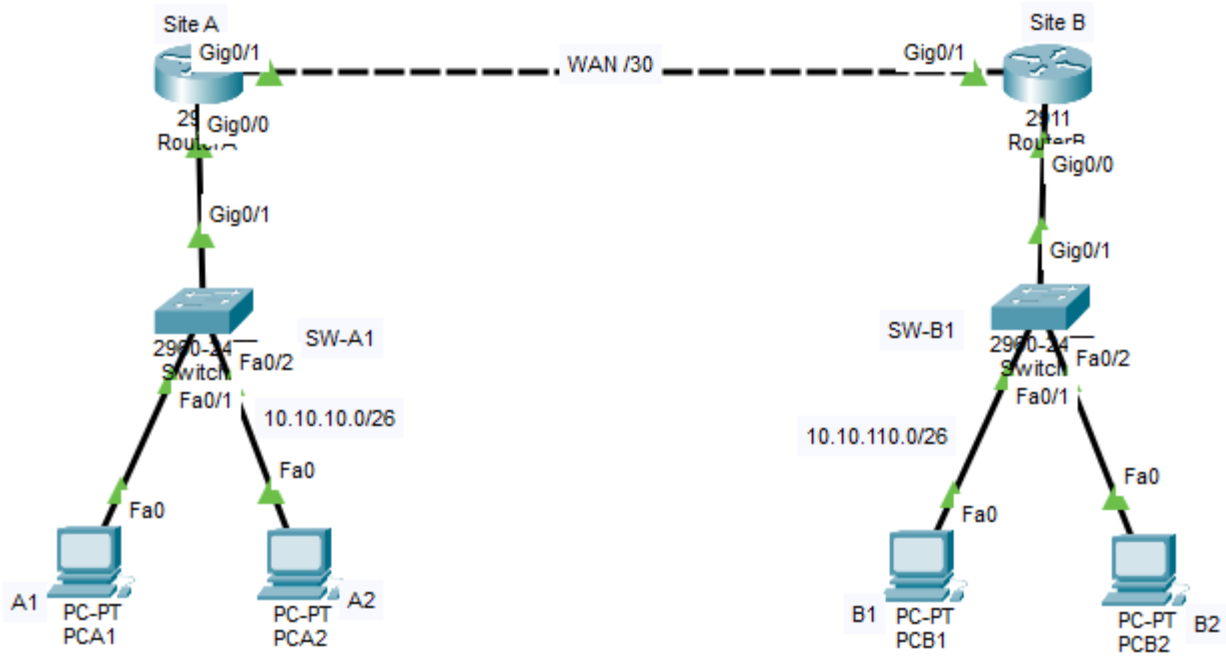
This Phase 2 prototype translates the CONNECTA Phase 1 network proposal into a small, testable IPv4 topology. Two enterprise sites are joined through a routed point-to-point link. Each site initially uses one active user LAN with router-based DHCP; dedicated addresses for application services, laboratory equipment and management remain reserved for future VLAN implementation. This staged approach supports local reachability, DHCP and intersite routing without claiming completion of VLAN tagging or advanced switching.

2. Updated network architecture

2.1 Working prototype versus target architecture

Component	Phase 2 prototype (proposed)	Later phase
Routers	R-A and R-B: Cisco Packet Tracer routers	OpenWrt on approved hardware
User LANs	One flat, active user segment per site	VLAN 10 with 802.1Q access/trunks
Switches	SW-A1 and SW-B1: basic Layer-2 access	Second managed switch at each site, LACP, RSTP
WAN	10.10.255.0/30 direct inter-router link	OSPF intersite adjacency
DHCP	Router DHCP on active user LANs	VLAN-specific DHCP
Server/services	Reserved Site A network; optional test server only if built	Go, FastAPI and persistent storage
Security/Internet	No production security claims	NAT, firewall policy, secure transport

Figure 1. Initial Cisco Packet Tracer topology connecting Site A and Site B.



2.2 Connection map and interfaces

The following interface names assume Cisco 2911-type routers with GigabitEthernet0/0 and GigabitEthernet0/1. Use straight-through Ethernet cables for router–switch and switch–PC links, and an appropriate link between routers.

From	Interface	To	Interface	Network
R-A	G0/0	SW-A1	G0/1	10.10.10.0/26
SW-A1	F0/1	PC-A1	FastEthernet0	10.10.10.0/26
SW-A1	F0/2	PC-A2	FastEthernet0	10.10.10.0/26
R-A	G0/1	R-B	G0/1	10.10.255.0/30
R-B	G0/0	SW-B1	G0/1	10.10.110.0/26
SW-B1	F0/1	PC-B1	FastEthernet0	10.10.110.0/26
SW-B1	F0/2	PC-B2	FastEthernet0	10.10.110.0/26

2.3 Design changes and justification

Compared with Phase 1, the immediate prototype intentionally reduces the topology to one access switch and one user LAN per site so that IPv4 addressing, DHCP and Layer-3 communication can be tested first. The earlier target architecture still calls for two managed switches per site, VLANs 10/20/30/99, trunks, inter-switch LACP, RSTP, OSPF and independent Internet access. The gateway addresses and reserved subnet blocks remain unchanged, which reduces migration rework.

3. VLSM IPv4 addressing plan

Parent block: 10.10.0.0/16. Each site keeps its own nonoverlapping network even when VLAN identifiers are reused. User host estimates come from Phase 1; capacities for other VLANs are planning reservations rather than a measured inventory.

Site/VLAN	Purpose	Host needed	Network	Mask	Usable range	Broadcast	Gateway	Allocation
A/10	Users	40	10.10.10.0/26	255.255.255.192	10.10.10.1 – 10.10.10.62	10.10.10.63	10.10.10.1	DHCP (active)
A/20	Go/FastAPI/database	Up to 30	10.10.20.0/27	255.255.255.224	10.10.20.1 – 10.10.20.30	10.10.20.31	10.10.20.1	Static (reserved)
A/30	Testing/lab	Up to 30	10.10.30.0/27	255.255.255.224	10.10.30.1 – 10.10.30.30	10.10.30.31	10.10.30.1	Reserved
A/99	Management	Up	10.10.99.0	255.255.25	10.10.99	10.10.99	10.10.99	Static

		to 14	/28	5.240	.1 – 10.10.99 .14	.15	.1	(reserv ed)
B/10	Users	25	10.10.110. 0/26	255.255.25 5.192	10.10.11 0.1 – 10.10.11 0.62	10.10.11 0.63	10.10.11 0.1	DHCP (active)
B/20	Future services	Up to 30	10.10.120. 0/27	255.255.25 5.224	10.10.12 0.1 – 10.10.12 0.30	10.10.12 0.31	10.10.12 0.1	Reserv ed
B/30	Testing/lab	Up to 30	10.10.130. 0/27	255.255.25 5.224	10.10.13 0.1 – 10.10.13 0.30	10.10.13 0.31	10.10.13 0.1	Reserv ed
B/99	Management	Up to 14	10.10.199. 0/28	255.255.25 5.240	10.10.19 9.1 – 10.10.19 9.14	10.10.19 9.15	10.10.19 9.1	Static (reserv ed)
WAN/ —	Inter-router	2	10.10.255. 0/30	255.255.25 5.252	10.10.25 5.1 – 10.10.25 5.2	10.10.25 5.3	A: 10.10.25 5.1; B: 10.10.25 5.2	Static (active)

Subnet capacity check: /26 provides 62 usable hosts, /27 provides 30, /28 provides 14 and /30 provides 2. These are distinct prefixes within 10.10.0.0/16.

4. Device and interface configuration

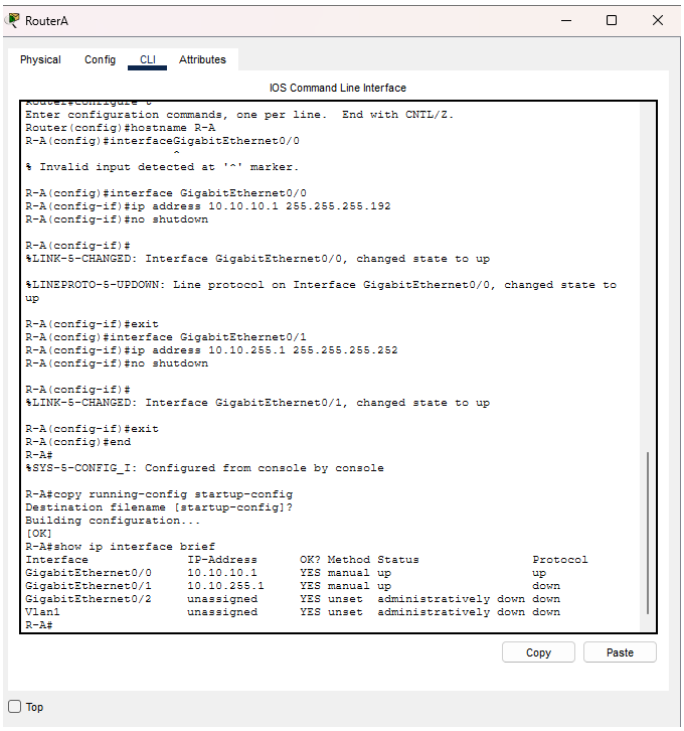
The initial network prototype was configured and tested using Cisco Packet Tracer. The following commands document the IPv4 addressing, DHCP configuration, static routing, and basic switch settings used during implementation. Configuration and connectivity evidence is presented in the corresponding figures.

4.1 Router R-A

```
enable
configure terminal
hostname R-A
interface GigabitEthernet0/0
 description SITE_A_USERS_TO_SW-A1
 ip address 10.10.10.1 255.255.255.192
 no shutdown
exit
interface GigabitEthernet0/1
 description INTERSITE_TO_R-B
```

```
 ip address 10.10.255.1 255.255.255.252
 no shutdown
exit
ip dhcp excluded-address 10.10.10.1 10.10.10.9
ip dhcp pool SITE_A_USERS
 network 10.10.10.0 255.255.255.192
 default-router 10.10.10.1
 dns-server 1.1.1.1
exit
ip route 10.10.110.0 255.255.255.192 10.10.255.2
end
copy running-config startup-config
```

Figure 2. Router R-A IPv4 interface configuration and verification.



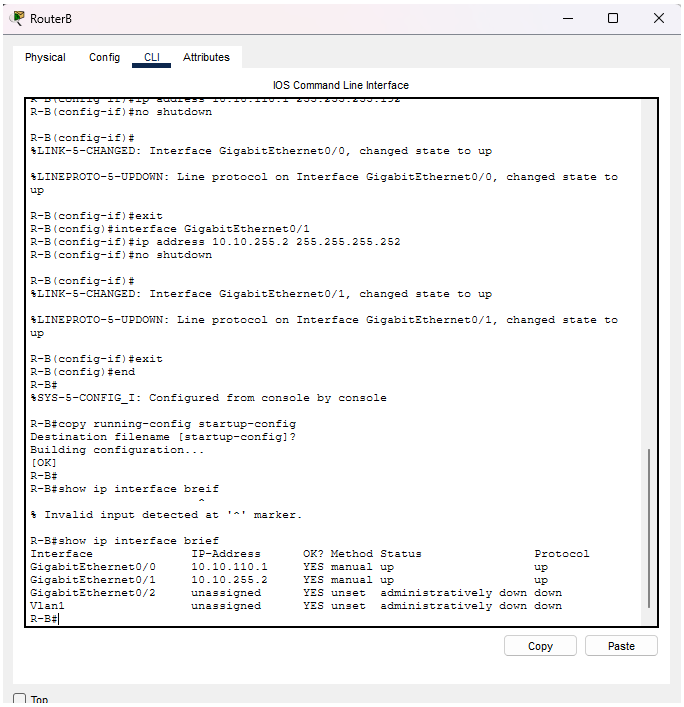
Router R-A was configured with IP address 10.10.10.1/26 on GigabitEthernet0/0 and 10.10.255.1/30 on GigabitEthernet0/1. The configuration was successfully saved. The LAN interface is up/up, while the intersite interface is currently up/down, pending verification of Router R-B.

4.2 Router R-B

```
enable
configure terminal
hostname R-B
interface GigabitEthernet0/0
 description SITE_B_USERS_TO_SW-B1
 ip address 10.10.110.1 255.255.255.192
 no shutdown
```

```
exit
interface GigabitEthernet0/1
 description INTERSITE_TO_R-A
 ip address 10.10.255.2 255.255.255.252
 no shutdown
exit
ip dhcp excluded-address 10.10.110.1 10.10.110.9
ip dhcp pool SITE_B_USERS
 network 10.10.110.0 255.255.255.192
 default-router 10.10.110.1
 dns-server 1.1.1.1
exit
ip route 10.10.10.0 255.255.255.192 10.10.255.1
end
copy running-config startup-config
```

Figure 3. Router R-B IPv4 interface configuration and verification.



Router R-B was configured with IPv4 address 10.10.110.1/26 on GigabitEthernet0/0 and 10.10.255.2/30 on GigabitEthernet0/1. Both interfaces were verified as operational (up/up), and the configuration was successfully saved.

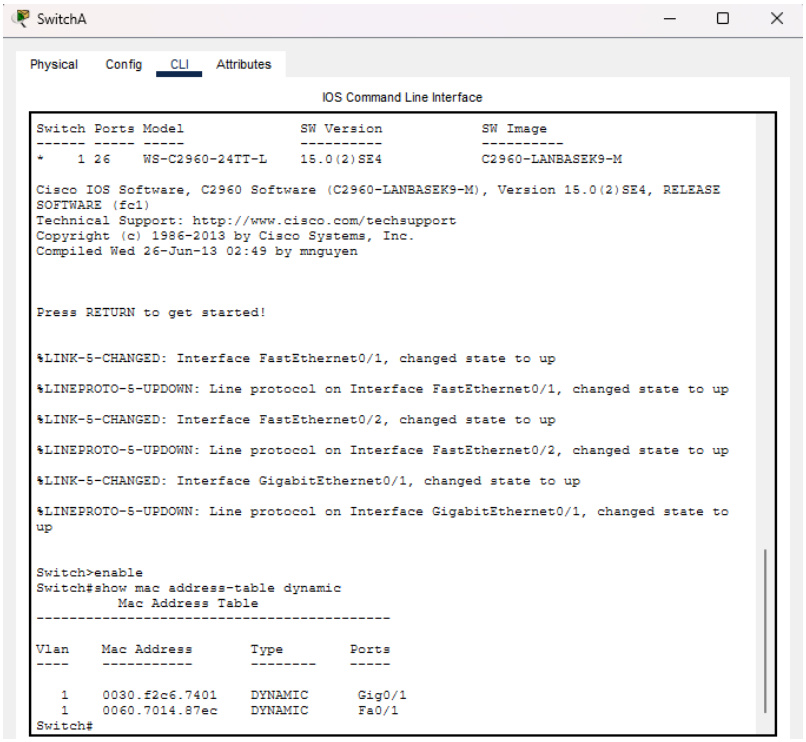
4.3 Basic Layer-2 switch configuration (repeat for each site)

```
enable
configure terminal
hostname SW-A1
interface GigabitEthernet0/1
```

```
 description UPLINK_TO_R-A
 switchport mode access
 no shutdown
exit
interface range FastEthernet0/1 - 2
 description USER_CLIENTS
 switchport mode access
 spanning-tree portfast
 no shutdown
exit
end
copy running-config startup-config
```

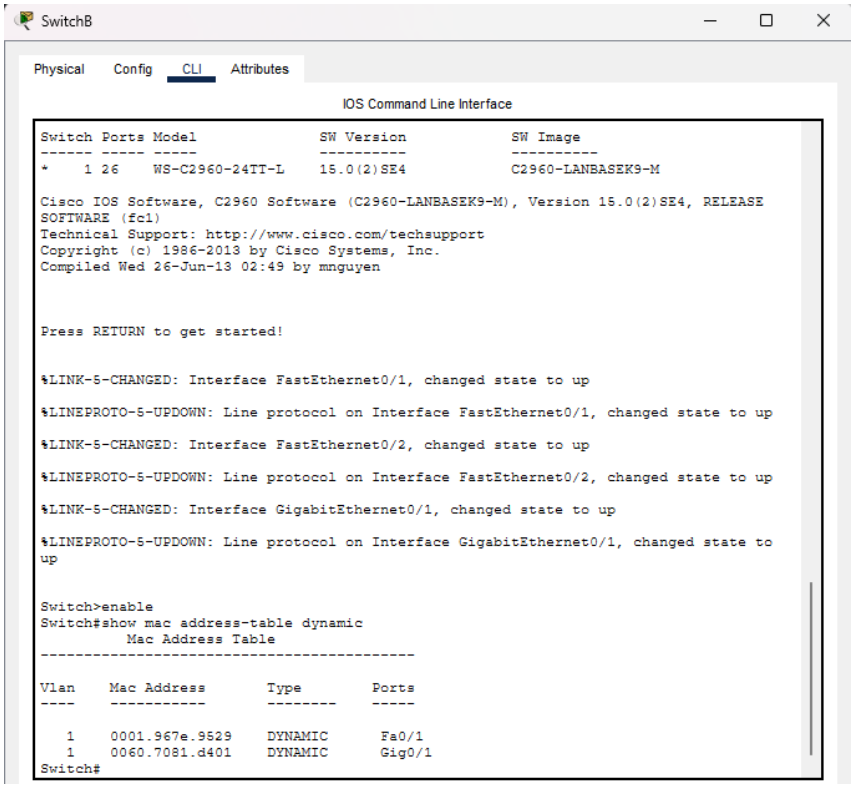
For the Site B switch, change the hostname to SW-B1 and the uplink description to UPLINK_TO_R-B. VLAN 10 is not created or activated in this temporary flat-LAN configuration; it is the planned VLAN identifier for Phase 3. A secure VTY/SSH configuration can be added only after checking feature support and establishing credentials outside shared files; no credentials are included here.

Figure 4. SW-A1 Dynamic MAC Address Table Verification.



The show mac address-table dynamic command confirms that SW-A1 successfully learned two dynamic MAC addresses through interfaces GigabitEthernet0/1 and FastEthernet0/1. Both entries belong to VLAN 1, which is currently used in the initial network prototype.

Figure 5. SW-B1 Dynamic MAC Address Table Verification.



The show mac address-table dynamic command confirms that SW-B1 successfully learned two dynamic MAC addresses through interfaces FastEthernet0/1 and GigabitEthernet0/1. Both entries belong to VLAN 1, which is currently used in the initial network prototype before VLAN segmentation is implemented.

4.4 Useful verification commands

Router and switch checks

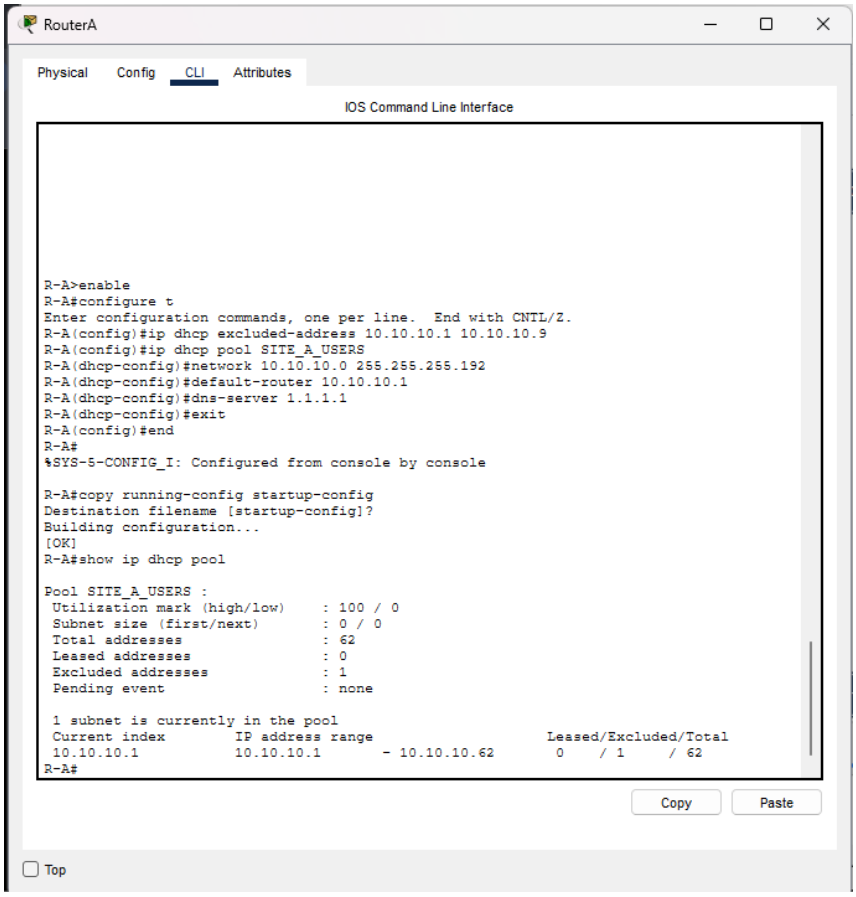
show ip interface brief
show running-config
show startup-config
show ip route
show ip dhcp pool
show ip dhcp binding
show interfaces status
show mac address-table dynamic
show interfaces gigabitEthernet0/0

Commands vary by router/switch model. On a switch, the dynamic MAC address table should learn each connected client MAC on its corresponding access port, and the router interface MAC on the uplink port once frames are exchanged. Record actual entries and ports, rather than inventing values.

5. DHCP implementation and verification

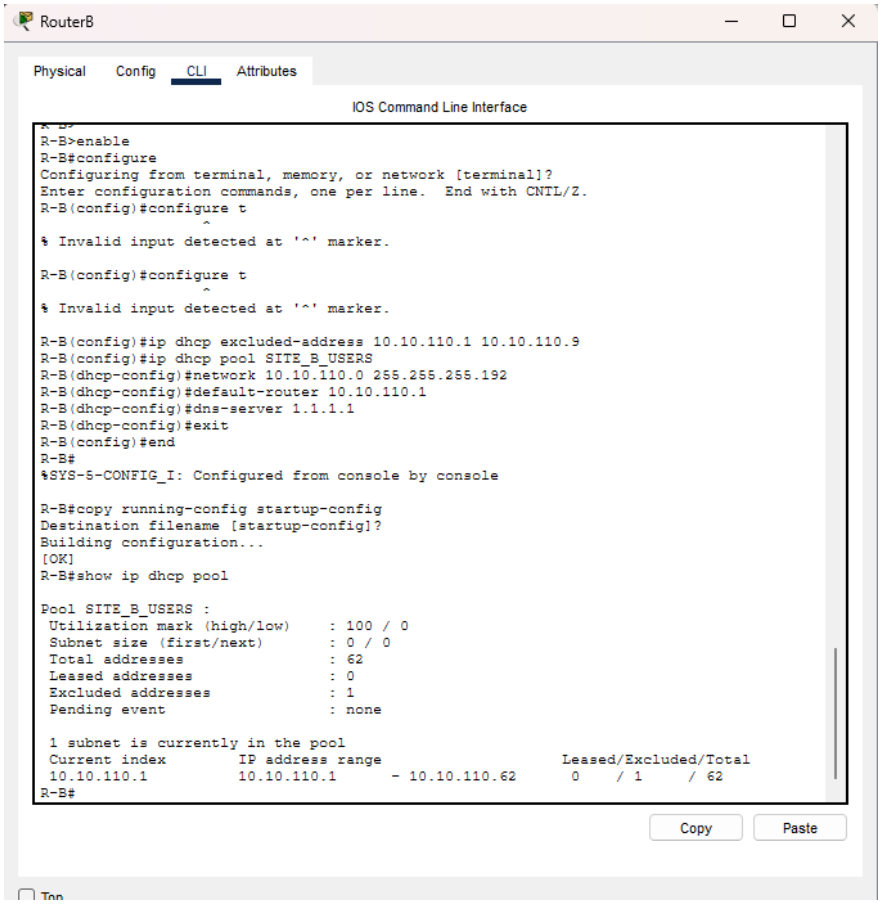
Pool	Subnet	Exclusion	Default gateway	Expected client range	Status
SITE_A_USERS	10.10.10.0/26	10.10.10.1–10.10.10.9	10.10.10.1	10.10.10.10–10.10.10.62	PASS Figures 6, 8, and 9
SITE_B_USERS	10.10.110.0/26	10.10.110.1–10.10.110.9	10.10.110.1	10.10.110.10–10.10.110.62	PASS Figures 7, 10, and 11

Figure 6. DHCP pool configuration on Router R-A.



The SITE_A_USERS DHCP pool was successfully configured on Router R-A for the 10.10.10.0/26 network. The gateway address is 10.10.10.1, and addresses from 10.10.10.1 to 10.10.10.9 are excluded from DHCP allocation. At this stage, no client addresses have been leased.

Figure 7. DHCP pool configuration on Router R-B.

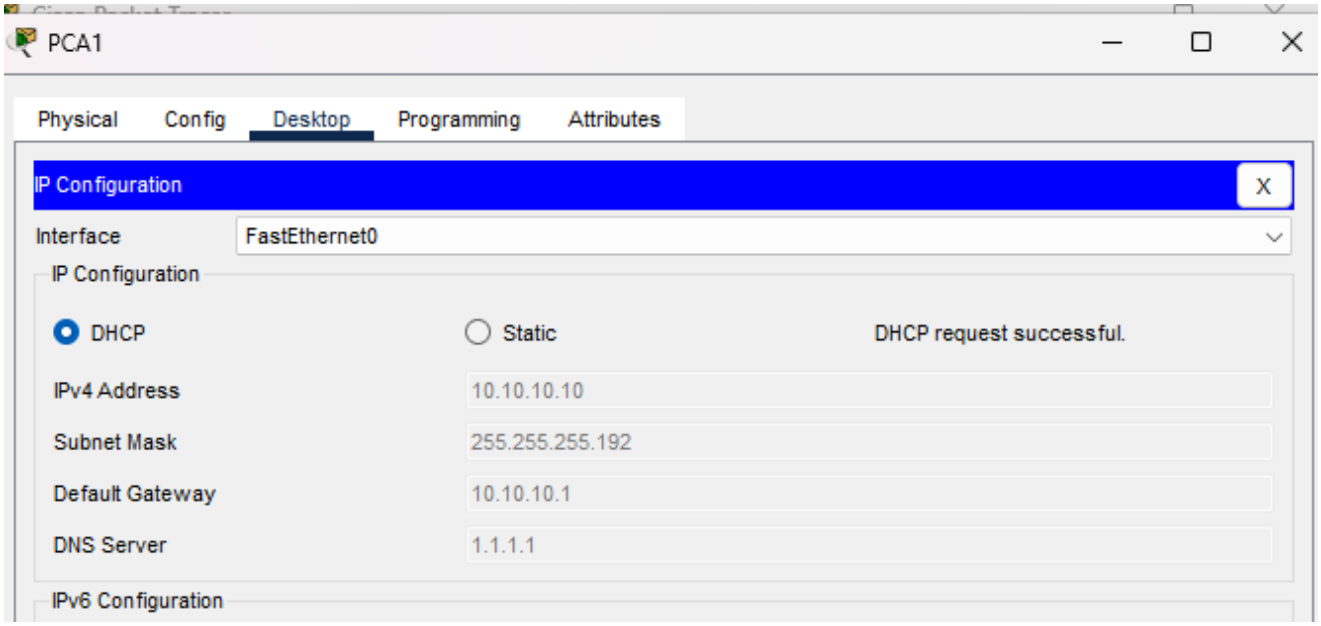


The SITE_B_USERS DHCP pool was successfully configured on Router R-B for the 10.10.110.0/26 network. Addresses from 10.10.110.1 to 10.10.110.9 were excluded from DHCP allocation, and the default gateway was configured as 10.10.110.1. No client addresses had been leased at the time of verification.

On each PC, select Desktop → IP Configuration → DHCP. Record the address, subnet mask and gateway that Packet Tracer assigns. The 1.1.1.1 DNS value is an illustrative resolver option; this isolated laboratory does not establish external DNS reachability.

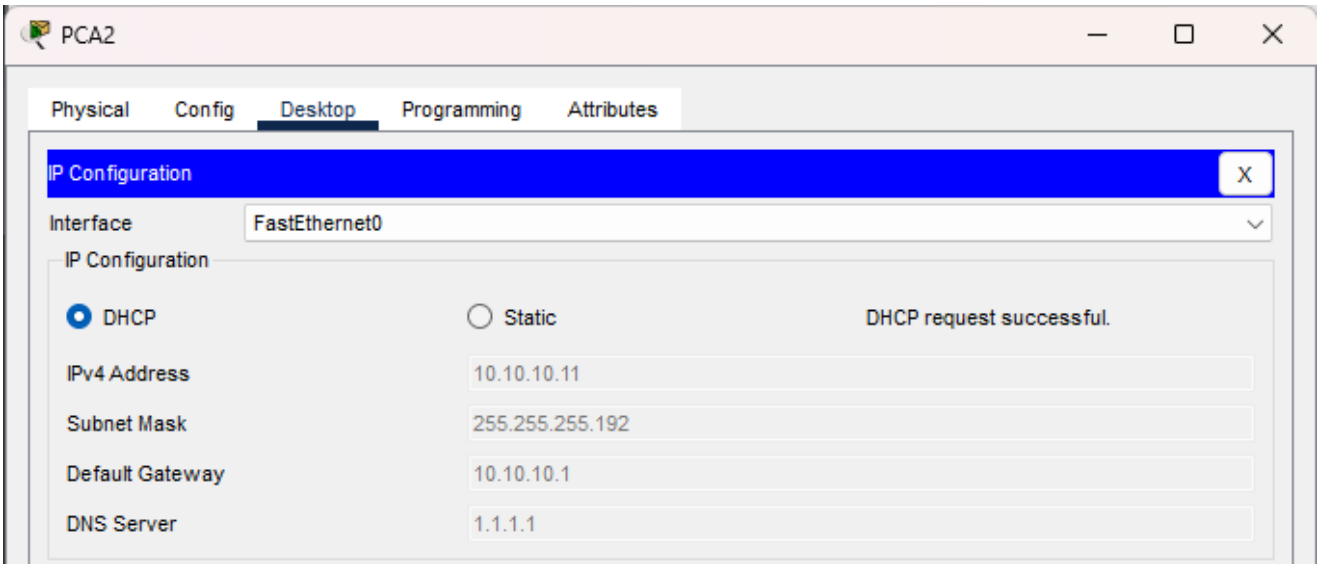
Client	Assigned IPv4	Mask	Gateway	Evidence
PC-A1	10.10.10.10	255.255.255.192	10.10.10.1	Figure 8
PC-A2	10.10.10.11	255.255.255.192	10.10.10.1	Figure 9
PC-B1	10.10.110.10	255.255.255.192	10.10.110.1	Figure 10
PC-B2	10.10.110.11	255.255.255.192	10.10.110.1	Figure 11

Figure 8. Successful DHCP address assignment on PC-A1.



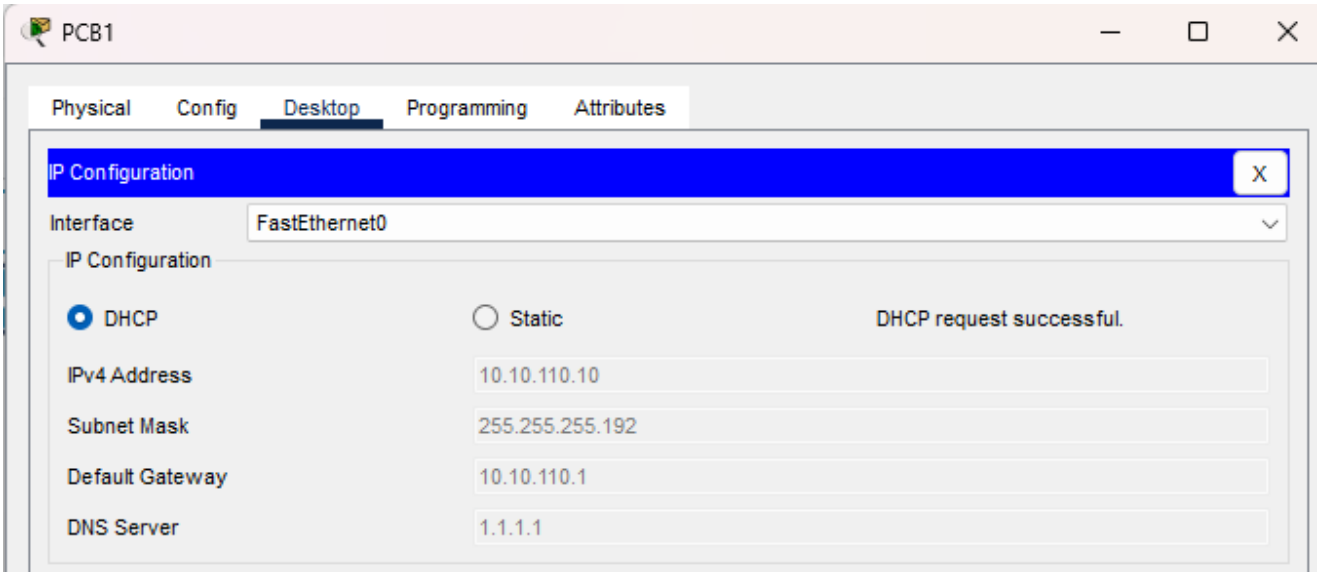
PC-A1 successfully obtained the IPv4 address 10.10.10.10 through DHCP, with subnet mask 255.255.255.192, default gateway 10.10.10.1, and DNS server 1.1.1.1. This confirms that DHCP address assignment is functioning correctly for this client in Site A.

Figure 9. Successful DHCP address assignment on PC-A2.



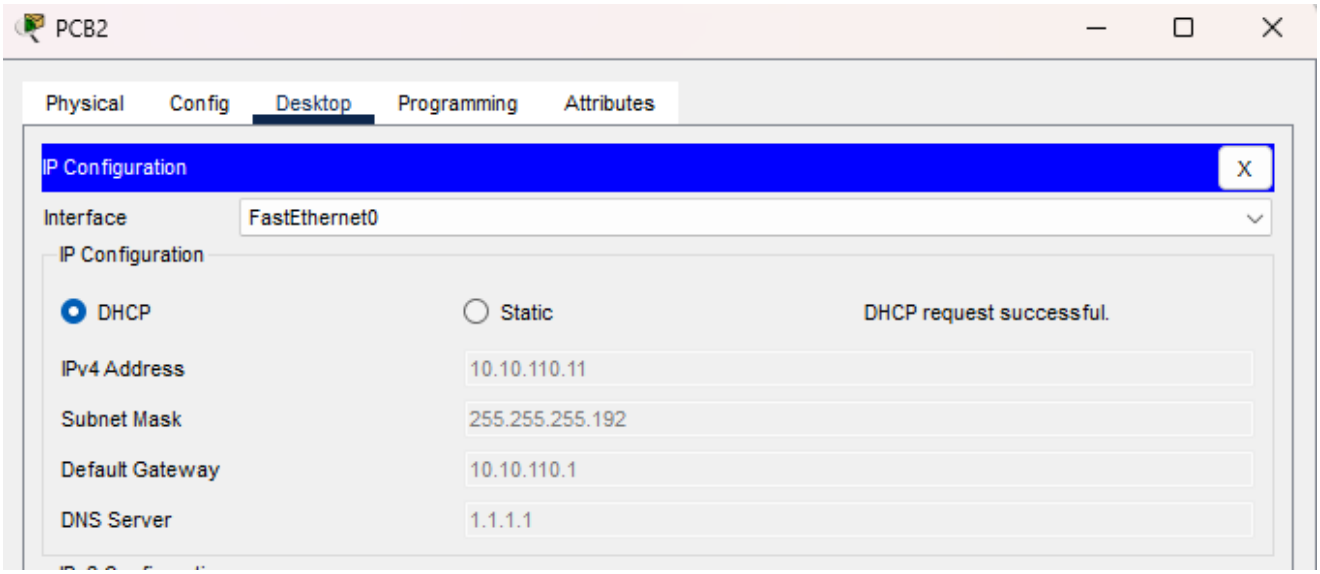
PC-A2 successfully obtained the IPv4 address 10.10.10.11 through DHCP, with subnet mask 255.255.255.192 and default gateway 10.10.10.1. This confirms that both test clients in Site A received valid DHCP configurations.

Figure 10. Successful DHCP address assignment on PC-B1.



PC-B1 successfully obtained the IPv4 address 10.10.110.10 through DHCP, with subnet mask 255.255.255.192, default gateway 10.10.110.1, and DNS server 1.1.1.1. This confirms that the DHCP service is functioning correctly for this client in Site B.

Figure 11. Successful DHCP address assignment on PC-B2.



PC-B2 successfully obtained the IPv4 address 10.10.110.11 through DHCP, with subnet mask 255.255.255.192, default gateway 10.10.110.1, and DNS server 1.1.1.1. This confirms that both test clients in Site B received valid DHCP configurations.

6. Initial routing and packet paths

During the initial Phase 2 implementation, Router R-A and Router R-B were configured with IPv4 addresses on their LAN and WAN interfaces. The intersite WAN network uses 10.10.255.0/30, with 10.10.255.1 assigned to R-A and 10.10.255.2 assigned to R-B.

Initial connectivity was verified using the ping command from R-A to R-B. The first test achieved an 80% success rate (4/5 packets), while the second test achieved 100% (5/5 packets), confirming successful communication across the directly connected WAN link.

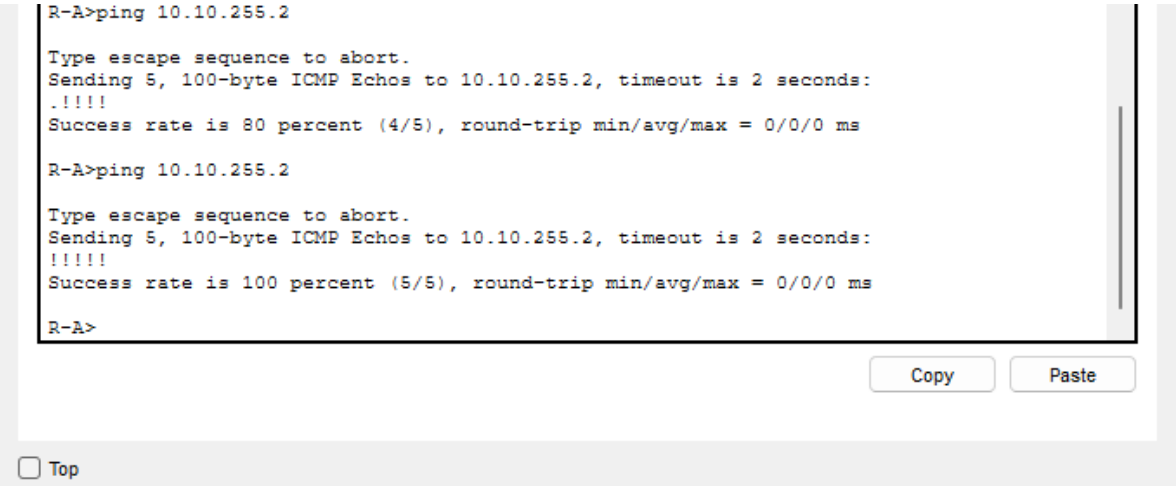
Static routing was successfully implemented on both routers. Router R-A uses the next-hop address 10.10.255.2 to reach the Site B network (10.10.110.0/26), while Router R-B uses 10.10.255.1 to reach the Site A network (10.10.10.0/26).

The routing tables were verified using the show ip route command, as demonstrated in Figures 13 and 14.

Intersite communication was successfully tested between PC-A1 and PC-B1 in both directions. The results are documented in Figures 15 and 16, confirming that both user networks can communicate through the configured static routes.

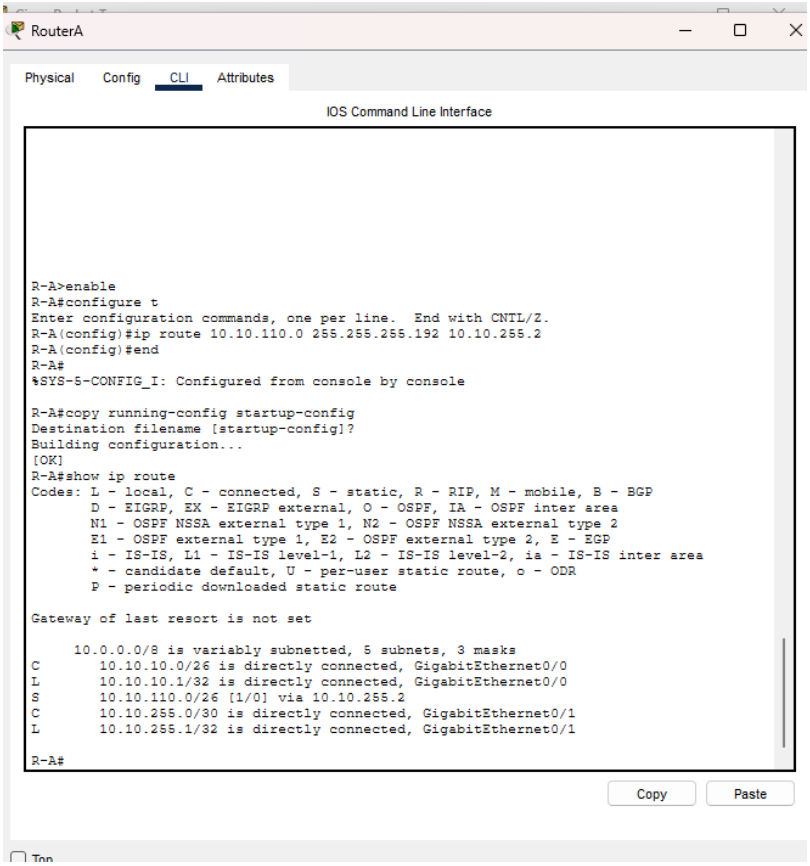
In Phase 3, these temporary static routes are planned to be replaced with OSPF.

Figure 12. Successful intersite connectivity verification.



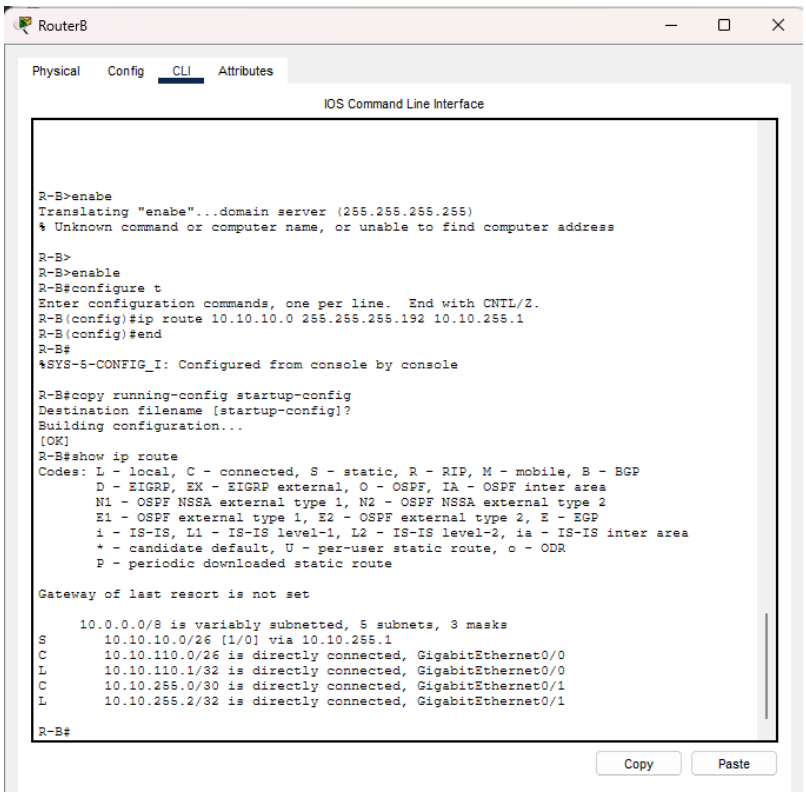
The connectivity test between Router R-A and Router R-B was repeated, achieving a 100% success rate (5 out of 5 packets received). This confirms successful IPv4 communication across the directly connected intersite WAN link.

Figure 13. Static routing table verification on Router R-A.



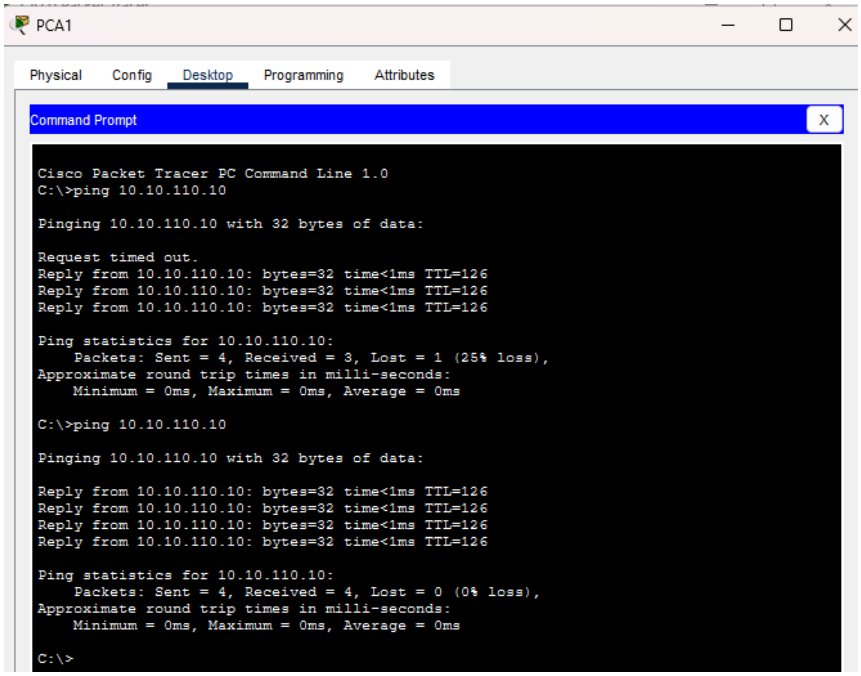
Router R-A was configured with a static route to the remote Site B user network (10.10.110.0/26) through next-hop address 10.10.255.2. The routing table confirms the static route with the code S and identifies the directly connected networks with the code C.

Figure 14. Static routing table verification on Router R-B.



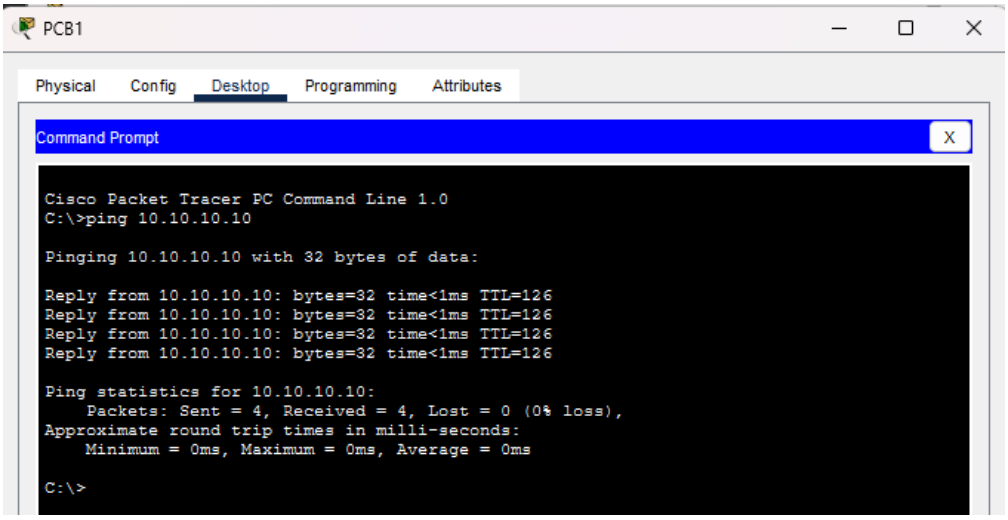
Router R-B was configured with a static route to the remote Site A user network (10.10.10.0/26) through next-hop address 10.10.255.1. The routing table confirms the static route with the code S and identifies the directly connected networks with the code C.

Figure 15. Successful intersite communication between PC-A1 and PC-B1.



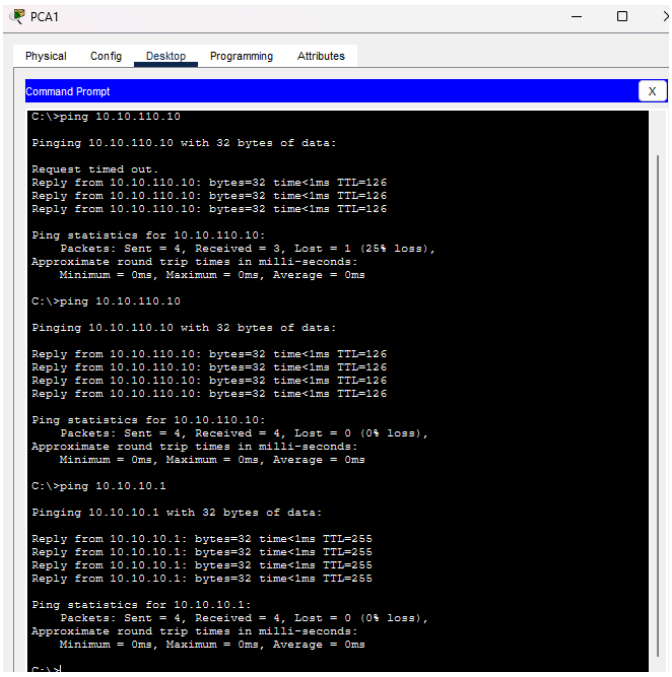
An intersite connectivity test was performed from PC-A1 (10.10.10.10) at Site A to PC-B1 (10.10.110.10) at Site B. The initial ping achieved a 75% success rate (3 out of 4 packets received). After repeating the test, all four packets were successfully received, achieving a 100% success rate. This confirms successful communication between the two user networks through the configured static routes.

Figure 16. Successful reverse intersite connectivity test from PC-B1 to PC-A1.



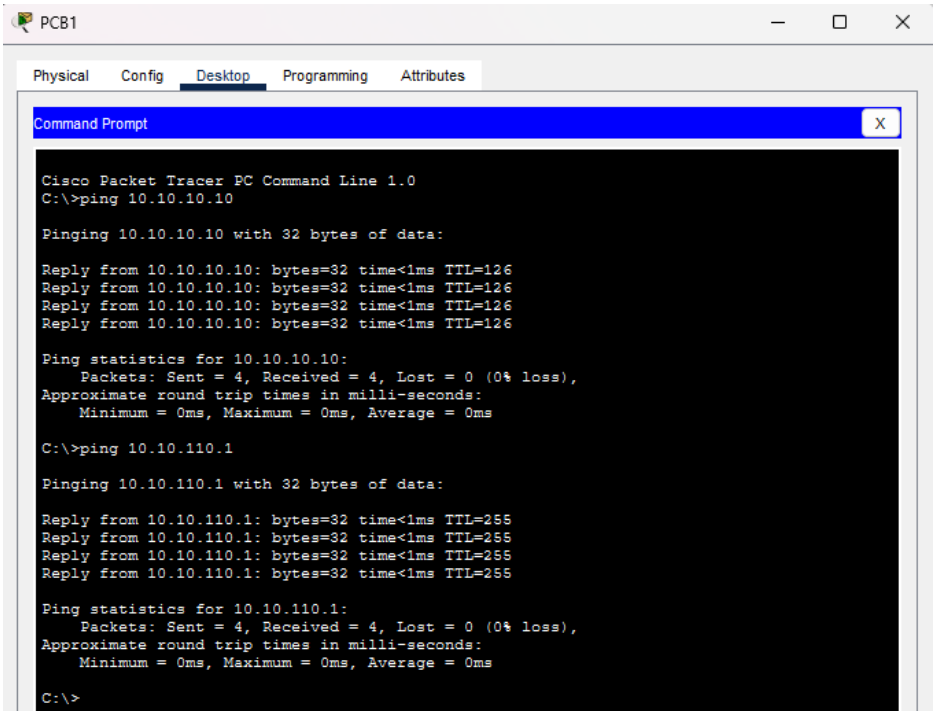
A reverse connectivity test was performed from PC-B1 (10.10.110.10) at Site B to PC-A1 (10.10.10.10) at Site A. All four ICMP packets were successfully received, with 0% packet loss. This confirms bidirectional IPv4 connectivity between the implemented user networks through the static routes configured on both routers.

Figure 17. Successful Local Connectivity Test from PC-A1 to Router R-A.



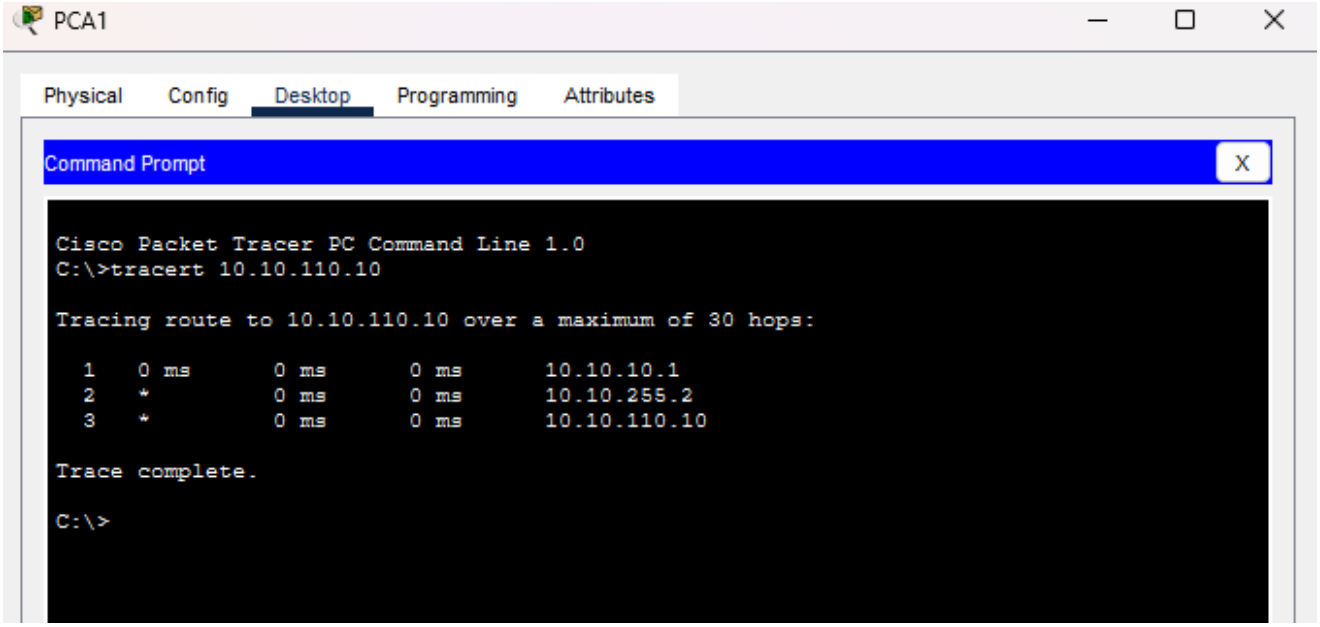
A local connectivity test was performed from PC-A1 (10.10.10.10) to its default gateway, Router R-A (10.10.10.1). All four ICMP packets were successfully received, with 0% packet loss. This confirms successful local IPv4 connectivity within Site A.

Figure 18. Successful Local Connectivity Test from PC-B1 to Router R-B.



A local connectivity test was performed from PC-B1 (10.10.110.10) to its default gateway, Router R-B (10.10.110.1). All four ICMP packets were successfully received, with 0% packet loss. This confirms successful local IPv4 connectivity within Site B.

Figure 19. Intersite Packet Path Verification from PC-A1 to PC-B1.



The expected communication path between PC-A1 and PC-B1 passes through SW-A1, Router R-A, Router R-B, and SW-B1. Static routes enable communication between the two user networks through the intersite WAN connection

7. Network administration and troubleshooting

Keep all access credentials, enable secrets, private keys and tokens out of the document and public repository. Save the .pkt project and configuration backups after verifying interface state. For SSH-capable Packet Tracer devices, plan local authentication and SSH rather than Telnet; the exact IOS commands depend on the model and software features.

7.1 Actual troubleshooting case

During the initial connectivity test between PC-A1 and PC-B1, the first ping attempt experienced 25% packet loss, receiving three out of four ICMP replies.

A possible explanation is the initial Address Resolution Protocol (ARP) process. Before sending packets to a remote network, the source device must resolve the MAC address of its default gateway. This initial address resolution can delay the first ICMP exchange.

The connectivity test was repeated without changing the network configuration. On the second attempt, all four packets were successfully received, resulting in 0% packet loss.

This result is consistent with temporary ARP resolution during initial communication, although the precise cause was not independently confirmed. The successful retest demonstrated that intersite connectivity was operational.

Field	Document the real observation
Issue ID	ISSUE-01
Symptom	Initial packet loss during the intersite ping test between PC-A1 and PC-B1.
Diagnosis	The initial test received 3 out of 4 ICMP replies. The precise cause of the packet loss was not confirmed.
Correction	The ping test was repeated without changing the network configuration.
Retest	Ping from PC-A1 to 10.10.110.10.
Result	Initial test: 75% success. Retest: 100% success, with 0% packet loss.
Evidence	Figure 15. Successful intersite communication between PC-A1 and PC-B1.

8. Preliminary test matrix

ID	Source	Destination	Procedure	Expected result	Observed / evidence	Status
T-01	PC-A1	R-A G0/0	ping 10.10.10.1	Replies received	Figure 17	PASS
T-02	PC-B1	R-B G0/0	ping 10.10.110.1	Replies received	Figure 18	PASS
T-03	PC-A1	DHCP R-A	Request DHCP lease	10.10.10.10–.62 /26, gateway .1	Figure 8	PASS
T-04	PC-B1	DHCP R-B	Request DHCP lease	10.10.110.10–.62 /26, gateway .1	Figure 10	PASS
T-05	R-A	R-B G0/1	ping 10.10.255.2	Replies received	Figure 12	PASS
T-06	PC-A1	PC-B1	ping assigned PC-B1 address	Replies received	Figure 15	PASS
T-07	PC-B1	PC-A1	ping assigned PC-A1 address	Replies received	Figure 16	PASS
T-08	R-A/R-B	Routing tables	show ip route	Connected + static routes	Figure 13-14	PASS

				shown		
T-09	SW-A1/SW-B1	MAC tables	show mac address-table dynamic	Learned entries match active ports	Figure 4 and 5	PASS
T-10	Site A VLANs 20/30/99	VLAN/trunk	Future simulation	Isolated segmentation	Planned for phase 3	PENDING
T-11	R-A and R-B	OSPF	Future neighbor verification	OSPF adjacency	Planned for phase 3	PENDING
T-12	Internet/WAN	NAT/firewall	Future controlled test	Permitted/blocked traffic	Planned for later implementation	PENDING

The preliminary tests confirmed successful local connectivity, DHCP address assignment, static routing, and bidirectional communication between Site A and Site B. The dynamic MAC address tables were successfully verified on both switches, SW-A1 and SW-B1. All nine preliminary tests (T-01 through T-09) were completed successfully. Advanced features, including VLAN segmentation, OSPF, NAT, and firewall policies, remain pending for subsequent implementation phases.

9. Pending work and migration to OpenWrt

Feature	Next action	Suggested owner	Planned checkpoint
VLANs 10/20/30/99	Create VLANs and assign access ports	Camila Lascano	Week 11
802.1Q / Router-on-a-Stick	Configure trunk and routed VLAN subinterfaces	Ashley Espinoza	Week 11
EtherChannel / RSTP	Add second switch per site; test LACP and loop recovery	Paula Simbaña	Week 11
OSPF	Replace temporary static routes, verify neighbors and routes	Camila Lascano	Week 11
DHCP per VLAN	Create VLAN-specific DHCP scopes and exclusions	Ashley Espinoza	Week 11
NAT and firewall	Define outbound NAT and least-privilege inter-VLAN policies	Paula Simbaña	Week 11–12
Go/FastAPI integration	Connect approved clients to hosted application services	Team	Weeks 12–15

OpenWrt hardware	Map interface names and migrate tested policy concepts	Team	From Week 11

9.1 Cisco-to-OpenWrt mapping

Cisco / Packet Tracer concept	OpenWrt approach (to validate on hardware)
GigabitEthernet interface / ip address	Configure physical interfaces and logical networks using UCI / /etc/config/network
802.1Q subinterfaces / Router-on-a-Stick	802.1Q device and bridge-VLAN configuration (hardware dependent)
ip dhcp pool and excluded-address	dnsmasq DHCP ranges, static leases and reserved addresses
ip route / OSPF	UCI static routes; OSPF via an available routing daemon (e.g. FRR), subject to image and package support
switchport / EtherChannel / RSTP	Managed-switch configuration or Linux bonding/bridging as supported; verify physical topology
ip nat / access lists	OpenWrt firewall4/nftables zones, forwarding policies and masquerading
IOS SSH management	OpenWrt SSH service and firewall-limited management access

9.2 Team Contributions

The CONNECTA Phase 2 network prototype was organized into three main areas of responsibility: network design and addressing, device configuration and connectivity, and testing and technical documentation.

Team Member	Contribution
Camila Lascano	**Network Topology Design, Ipv4 Vlsm Addressing Plan, Subnet Allocation, And Identification Of Planned Vlans.**
Ashley Espinoza	**Cisco Router And Switch Configuration, Dhcp Implementation, Static Routing, And Local And Intersite Connectivity Testing.**
Paula Simbaña	**Verification Of Test Results, Mac Address Table Analysis, Troubleshooting Documentation, Preparation Of The Technical Report, And Repository Organization.**